CNC Router Overview

Overview

The CNC Router was my third major project and the first where I integrated mechanical design, electronics, and software into a single project. The goal was to create a computer-controlled cutting device capable of machining wood and plastic with decent accuracy, allowing me to level up the parts I could make for future projects as well as use it as an engraver for gifts and stationery.

Design

The design drew inspiration from a similar machine on Youtube by Ivan Miranda but was built entirely from reused parts and 3D-printed components. The frame was constructed from 2020 aluminum extrusions for rigidity, while the motion system relied on lead screws and linear rails and belts I harvested from my old Flashforge Creator Pro printer.

The CNC was controlled using 4 NEMA 17 stepper motors connected to their own motor microstepper drivers (the two Y axis motors were connected to the same driver but in reverse) and an Arduino UNO R3 running GRBL, which ran the Gcode for the machine. All the electronics were powered with a Meanwell LRS-350-24 power supply which could supply 24V at 14.6 A. I used the open-source Universal Gcode Sender to handle the calibration, homing and other general CNC controls. Since I had access to Fusion 360, I used the CAM software that comes with the student license to convert my 3D models into toolpaths.

The spindle motor of this machine was my readily available corded Dewalt handheld router meant for woodworking. Because the goal of this project was to cut wood and plastic, I did not need an expensive spindle with precise controls.

Challenges and Things Learned

The CNC Router was an eye opener in my engineering journey. I learned how important precision was when making precise machines. The skills from the basic electrical and software engineering will carry on to future projects. However, this project was not fully successful. After one run of the cnc, I noticed that the dust and chips from cutting was getting stuck on the belts and lead screws of the machine. This caused significant issues as the weaker NEMA 17s could not provide enough torque to keep the machine running. The fix for this would be to add a dust shield around the spindle motor that redirects all the chips and dust away from the machine.